

# Max Rakitin

## Experience

### About

Name: Max Rakitin (a.k.a. Maksim S. Rakitin)

Summary: I am a group leader of the Data Acquisition and Detectors group of the Data Science and Systems Integration Division of NSLS-II, BNL.

News: "Computer, Is My Experiment Finished?" (September 16, 2022)  
<https://www.bnl.gov/newsroom/news.php?a=220832>

"Seeing the Forest Through the Trees: Brookhaven Lab Scientists Develop New Computational Approach to Reduce Noise in X-ray Data." (April 18, 2022)  
<https://www.bnl.gov/newsroom/news.php?a=219533>

Links: [○ BNL](#) • [SBU](#) • [SUSU](#)  
[○ @mrakitin](#) • [in @mrakitin](#) • [Google Scholar](#) • [R<sup>®</sup> ResearchGate](#)  
[○ ID](#) ORCID: 0000-0003-3685-852X

### Experience



2024.06–present

**Computational Scientist (RS-5), Group Leader**, *Data Science and Systems Integration (DSSI) Division, Data Acquisition and Detectors Group, NSLS-II, Brookhaven National Laboratory, Upton, NY* (<https://www.bnl.gov>, [DSSI](#)).

**Supervisor:** Dr. Stuart Wilkins



2023.01–2024.06

**Computational Scientist (RS-5), Deputy Group Leader**, *Data Science and Systems Integration (DSSI) Program, Data Acquisition and Detectors Group, NSLS-II, Brookhaven National Laboratory, Upton, NY* (<https://www.bnl.gov>, [DSSI](#)).

**Supervisor:** Dr. Thomas Caswell



2021.03–2022.12

**Associate Computational Scientist (RS-4), Supervisor**, *Data Science and Systems Integration (DSSI) Program, NSLS-II, Brookhaven National Laboratory, Upton, NY* (<https://www.bnl.gov>, [DSSI](#)).

**Supervisor:** Dr. Stuart Campbell



2019.10–2021.03

**Associate Computational Scientist**, *Data Acquisition, Management and Analysis (DAMA) Group, NSLS-II, Brookhaven National Laboratory, Upton, NY* (<https://www.bnl.gov>).

**Supervisor:** Dr. Stuart Campbell

2017.11–2019.09



**Assistant Computational Scientist, Data Acquisition, Management and Analysis (DAMA) Group, NSLS-II, Brookhaven National Laboratory, Upton, NY**  
(<https://www.bnl.gov>).

**Supervisor:** Dr. Stuart Campbell

#### Software Projects:

- **Bluesky** — a library for experiment control and collection of scientific data and metadata, [docs](#), [GitHub](#).
- **Ophyd** — a device abstraction library, [docs](#), [GitHub](#).
- **Databroker** — a simple, user-friendly interface for retrieving stored data and metadata from multiple sources, [docs](#), [GitHub](#).
- **Sirepo-Bluesky** — an interface library between the **Bluesky** data acquisition framework and the **Sirepo** browser-based interface framework to beamline and accelerator simulation codes, [docs](#), [GitHub](#).
- **Blop** — a Bayesian optimization library for autonomous experiment control, [docs](#), [GitHub](#).

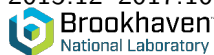
#### Funded Projects (PI roles):

- **LDRD Type B “Simulation-aided Instrument Optimization Using Artificial Intelligence and Machine Learning Methods”** (LDRD #22-031, FY2022–FY2024, Oct. 2021–Sep. 2023) — PI: M. Rakitin, co-PIs: Y. Du, M. Fedurin. BNL Laboratory-Directed Research and Development project, <https://www.osti.gov/biblio/1972474>.
- **DOE SBIR “X-ray Beamline Control with an Online Model for Automated Tuning and Reconfiguration”** (Award No. DE-SC0020593, 2020–2023) — PI at BNL (subcontract). Lead organization: RadiaSoft, LLC (PI: B. Nash).
- **DOE SBIR “An Integrated Platform for High Performance Data Management and Analysis at X-ray Light Sources”** (Award No. DE-SC0021553, 2021–2024) — PI at BNL (subcontract). Lead organization: RadiaSoft, LLC (PI: N. Cook).

#### Responsibilities:

- Scientific software development & maintenance.
- Packaging and maintenance of conda packages (conda-forge, and formerly lightsource2-tag and nsls2forge conda channels).
- Deployment of the software with Ansible.
- NSLS-II beamlines and users support.
- Active collaboration with peers at other US DOE National Labs’ User Facilities (SLAC — LCLS & SSRL, LBNL — ALS, ANL — APS, ORNL — SNS) and international facilities (DLS, PSI, BESSY-II, CLS, DESY).
- Teaching people to use our software stack.
- Mentor to student interns (summer 2016: 1 intern, summer 2018: 1 intern, summer 2019: 1 intern, spring and summer 2020: 2 interns, summer 2021: 2 interns, summer 2023: 1 intern, summer 2024: co-mentor to 1 intern).

2015.12–2017.10



**Research Associate (Postdoc), NSLS-II, Brookhaven National Laboratory, Upton, NY**  
(<https://www.bnl.gov>).

**Supervisor:** Dr. Oleg Tchoubar (Chubar)

**Software Projects:**

- **Synchrotron Radiation Workshop (SRW)** — computer code for X-ray source and optics simulations, <https://github.com/ochubar/SRW>.
- **Sirepo** — a cloud-based framework for SRW, <https://github.com/radiasoft/sirepo>.
- **Image processing and data visualization**, <https://github.com/mrakitin/plotting> and <https://github.com/mrakitin/experiments>.
- **CRL simulator** — a code for simulation of a translocator (compound refractive lenses (CRL) for X-ray focusing), <https://github.com/mrakitin/bnlcrl>.

**Responsibilities:**

- Adding new functionality to Sirepo (e.g., new optical elements, new reports, [implementation](#) of dynamical access of crystal data and optical constants from external servers).
- Adding new functionality to SRW (e.g., new optical elements).
- Creation of new SRW/Sirepo “virtual beamline” scripts/examples.
- Creation and update of [Sirepo](#) & [SRW](#) wiki documentation.
- SRW & Sirepo users community support via GitHub issues, email communication, *etc.*
- Deployment of SRW & Sirepo to NSLS-II servers for usage by beamline scientists.
- Computational support for NSLS-II beamlines — carrying out required SRW & Sirepo simulations.
- Assistance with scan plans preparation using **Bluesky** data collection framework and **ophyd** package for controlling motors, detectors, *etc.*
- Participation in X-ray experiments (SRX, ESM, SMI, CHX beamlines of NSLS-II) — carrying out measurements and bulk data analysis, processing & visualization using Python (NumPy, SciPy, Matplotlib, PIL, *etc.*; raw data and images, NumPy arrays, HDF5 datasets).
- Implementation and integration of samples simulation code for coherent scattering experiments in SRW and Sirepo (collaborative project with CFN).
- Assisting with development of the Hartmann mask optical element in SRW and its implementation in Sirepo (collaborative project with the Metrology group of NSLS-II).
- Tutorship of summer students.

2013.10–2015.12



**Postdoctoral Associate (Postdoc)**, *Prof. Oganov's lab, Department of Geosciences, Stony Brook University, Stony Brook, NY* (<https://stonybrook.edu>, <https://uspex-team.org/en/uspex>).

**Supervisor:** Prof. Artem R. Oganov

**Software Projects:**

- Refactoring of [USPEX](#) code and porting it from Matlab to Python using modern programming techniques.
- *Ab initio* investigation of BeF<sub>2</sub> and SiO<sub>2</sub> systems in a wide pressures range ([paper](#)) using [USPEX](#), [VASP](#), [Quantum Espresso](#), [Phonopy](#).
- Prediction of secondary structures of proteins from knowledge of sequences of amino acids. Development of USPEX interface for [Tinker](#).

**Responsibilities:**

- Main developer of the project on rewriting **USPEX** in Python with use of NumPy, SciPy, TkInter, etc. Web-based API creation.
- New **releases** of Matlab/Octave version (USPEX 9.4.1, 9.4.2, 9.4.3, 9.4.4), new features, bug fixes, maintenance.
- Development and maintenance of USPEX continuous integration system (automatic nightly builds + testing).
- Development of tests for USPEX interface with VASP, GULP, LAMMPS, CASTEP, Quantum Espresso, SIESTA, CP2K, QuantumWise ATK, DMACRYS, Tinker, FHI-aims.
- Development and maintenance of online utilities <https://uspex-team.org> using JavaScript, jQuery, PHP, HTML, JSON, REST API, WebGL, XML, SVN, interfaces to Python, Fortran, Matlab programs, etc.
- Installation, configuration, and maintenance of an instance of a Trac system and SVN.
- Creation and maintenance of **USPEX manual** in  $\text{\LaTeX}$ , PDF, HTML formats.
- **USPEX community** support.

2008.10–2013.09 **Researcher**, *Department of General and Theoretical Physics, South Ural State University, Chelyabinsk, Russia* (<https://www.susu.ru/en>).



**Supervisor:** Prof. Alexander A. Mirzoev

**Software Projects:**

- Performing research on investigating influence of hydrogen on structure and properties of iron-based alloys using DFT methods (**WIEN2k**).

**Responsibilities:**

- Development of bash/Python utilities for monitoring WIEN2k calculations in PBS/Torque and SLURM queues and for processing data of calculations, developing mail notification system for the calculations.
- Installation of WIEN2k on a PC and on the university supercomputers (SKIF Ural, SKIF Aurora, Tornado).
- Teaching and consulting students how to use WIEN2k both on a PC and on supercomputers.

2007.06–2013.10 **QA Engineer, QA Team Leader**, *Applied Technologies Ltd., Chelyabinsk, Russia* (<https://www.appliedtech.ru/en/>), a partner of Rocket Software Inc., USA (<https://www.rocketsoftware.com>)



**Supervisor:** Valery Ermakov, CEO

**Software Projects:**

- Tivoli Enterprise Portal (TEP) (August 2011 – October 2013) — QA team leader. Responsibilities: software testing on Windows, Linux, Linux on z, z/OS operating systems, GUI testing automation using [IBM Rational Functional Tester](#), creation of TEP Automated Testing System (TATS) for automatic data verification using Python, REST API, JSON, XML, PHP, HTML, JavaScript, jQuery in six TEP-enabled products:
  - [Tivoli Advanced Reporting and Management for DFSMSHsm](#);
  - [Tivoli Advanced Audit for DFSMSHsm](#);
  - [Tivoli Advanced Catalog Management for z/OS](#);
  - [Tivoli Advanced Backup and Recovery for z/OS](#);
  - [Tivoli Advanced Allocation Management](#);
  - [Tivoli Automated Tape Allocation Manager for z/OS](#).Regular voice and video conversations with colleagues from the US.
- [Tivoli Storage Manager for z/OS Media](#) (February 2011 – August 2011) — QA engineer. Responsibilities: software testing on Windows, Linux on z, Solaris, AIX, z/OS operating systems with IBM DB2 databases, testing automation using IBM-developed tools, bash, batch.
- [IBM Tivoli Advanced Allocation Management](#) (January 2009 – January 2011) — QA engineer. Responsibilities: software testing on z/OS, testing automation using bash, JCL, REXX. Regular voice and video conversations with colleagues from the US.
- [Rocket Servergraph Data Protection Expert](#) (June 2007 – December 2008) — QA engineer. Responsibilities: software testing on Windows, Linux, Solaris, HP-UX, AIX operating systems with PostgreSQL databases, GUI testing automation using Autolt3, automation of data verification using Perl and SQL. Regular voice and video conversations with colleagues from the US.

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2006–2007  **System administrator**, *Department of General and Theoretical Physics, South Ural State University, Chelyabinsk, Russia* (<https://www.susu.ru/en>)  
Administer, secure and support Windows and Linux systems. Support HTTP server (Apache), proxy server (Squid). Help others to troubleshoot operating system, software, hardware and other issues. Also support the testing system for students.